

SAMARTH SONI

AI Engineer | Machine Learning Engineer

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Experience

Q-Lab, UC San Diego — *Research Collaborator - Generative Modeling* Jul 2026 – Sep 2026

- Develop and evaluate PyTorch generative-modeling pipelines on ImageNet-1K using frozen VAE latent representations and distributed training on NVIDIA L40S GPUs.
- Design controlled experiments across AdamW and Muon optimizers, training objectives, auxiliary losses, timestep sampling, and MeanFlow/Drifting configurations to analyze generation quality, convergence, and stability.
- Prepared and validated 1.28 million training samples and implemented mixed-precision training, EMA, checkpointing, sampling, and Jacobian-vector-product computations.

Control Loop — *Software Engineer - AI Systems* Nov 2025 – Mar 2026

- Architected and built the company's first production Python/AWS AI document-processing system, handling approximately 10,000 insurance documents per month across OCR, LLM extraction, validation, summarization, and downstream routing.
- Built schema-driven workflows for policy metadata and operational deadlines with validation, idempotent processing, structured logging, monitoring, and failure handling.
- Evaluated and selected LLMs and prompting strategies based on extraction accuracy, reliability, latency, and inference cost, then integrated the chosen approach into production.

Jaypee Institute of Information Technology — *Faculty Fellow - Machine Learning* Jun 2025 – Sep 2025

- Developed end-to-end computer-vision and machine-learning pipelines for oral-cancer image classification, including preprocessing, feature extraction, supervised learning, hyperparameter tuning, and quantitative model evaluation.
- Conducted controlled experiments and statistical analysis to compare classifiers and improve predictive performance; co-authored resulting research published in *Scientific Reports*.
- Mentored students in Python and machine-learning laboratories and reviewed model implementations, experimental methodology, and technical results.

The Zen India — *AI Engineer Intern* Jan 2025 – May 2025

- Integrated speech-to-text and neural machine-translation models into a multilingual healthcare platform supporting patient intake and real-time clinical transcription.
- Built Python APIs for ML model serving and application integration with structured logging, error handling, and monitoring for reliable production AI workflows.

Selected Project

Deep Learning and NLP from Scratch — *Independent Project* 2026

- Implemented Word2Vec, GloVe, RNN, LSTM, GRU, Seq2Seq, Bahdanau and Luong attention, and Transformer attention architectures in PyTorch using custom datasets, training loops, evaluation pipelines, and inference workflows.

Technical Skills

Programming and Data: Python, SQL, C++, Bash, NumPy, pandas, data preprocessing, EDA, feature engineering

Machine Learning and Deep Learning: PyTorch, TensorFlow, scikit-learn, Transformers, computer vision, NLP, generative AI, statistical modeling, hyperparameter tuning

Generative AI and LLMs: LLMs, prompt engineering and evaluation, information extraction, text summarization, generative modeling, transformer architectures

ML Engineering: Training and inference pipelines, model serving, distributed GPU training, mixed precision, experiment design, monitoring, FastAPI, REST APIs, Docker, Linux, Git

Cloud: AWS Lambda, S3, SNS, Textract, Bedrock, SES

Education

Jaypee Institute of Information Technology — Noida, India
B.Tech. in Computer Science and Engineering; GPA: 8.7/10

May 2025

Selected Publications

- Soni, S. et al., "An Adaptive Multiscale Local Mesh Ternary Pattern Technique with Extensive Pre-processing and Grey Wolf Optimisation Based Classifiers for Oral Cancer Image Classification," *Scientific Reports*, 2026.
- Soni, S. et al., "Evaluating Deep Learning Approaches for Grammatical Error Correction: A Comparative Study of Seq2Seq Models," IEEE PDGC, 2024. Two additional peer-reviewed IEEE publications in NLP and medical imaging.